

SAULT STE MARIE, ON, Canada

WMO: 712600

Lat: 46.4840N

Lon: 84.4992W

Elev: 630

StdP: 14.36

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.3	-7.5	-21.7	1.7	-12.6	-15.6	2.4	-6.6	28.8	23.3	24.9	20.7	3.5	80	0.612

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	83.3	70.0	80.2	68.0	77.4	66.4	72.1	79.9	70.1	77.3	68.3	75.0	8.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.5	111.1	76.7	67.7	104.2	74.1	65.7	97.2	71.8	36.3	80.0	34.5	77.2	33.0	75.1	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.0	19.9	17.7	-22.0	88.0	7.3	2.5	-27.2	89.8	-31.5	91.3	-35.6	92.7	-40.8	94.6		
(5)			-22.4	75.8	7.1	2.0	-27.5	77.2	-31.6	78.4	-35.6	79.5	-40.8	81.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	41.3	15.0	16.3	24.7	37.7	50.3	59.8	64.8	64.2	57.9	45.9	34.5	23.3	(6)
(7)		DBStd	19.78	12.30	11.51	11.46	8.12	7.80	6.76	5.66	5.28	7.40	7.82	8.65	10.18	(7)
(8)		HDD50	4775	1086	943	785	375	93	6	0	0	15	178	468	826	(8)
(9)		HDD65	8813	1551	1363	1248	818	461	184	75	80	234	592	916	1291	(9)
(10)		CDD50	1612	0	0	2	7	101	299	458	439	252	52	2	0	(10)
(11)		CDD65	176	0	0	0	0	5	27	68	54	21	1	0	0	(11)
(12)		CDH74	1469	0	0	1	2	74	301	591	367	129	4	0	0	(12)
(13)		CDH80	275	0	0	0	0	13	59	127	57	19	0	0	0	(13)
(14)	Wind	WSAvg	7.6	7.9	7.9	8.0	8.5	8.0	6.9	6.4	6.2	6.5	8.0	8.6	8.6	(14)
(15)	Precipitation	PrecAvg	36.7	3.1	1.9	2.4	2.7	2.8	3.2	2.9	3.4	3.9	3.4	3.7	3.3	(15)
(16)		PrecMax	48.3	5.0	5.5	5.4	4.7	7.3	9.1	6.5	6.9	7.9	6.2	6.9	6.3	(16)
(17)		PrecMin	24.9	0.9	0.2	0.5	1.2	0.6	0.4	0.3	0.9	1.3	0.3	1.2	0.4	(17)
(18)		PrecStd	5.1	1.0	1.1	1.2	1.0	1.4	1.6	1.4	1.3	1.7	1.3	1.3	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.1	42.4	57.0	68.5	81.5	85.7	87.5	85.1	82.9	73.0	59.3	45.6	(19)
(20)			MCWB	39.2	39.0	48.9	54.4	66.7	71.3	72.2	71.7	70.5	64.0	53.0	43.6	(20)
(21)		2%	DB	36.8	37.8	48.5	61.1	74.9	81.0	83.5	81.2	77.6	66.7	52.7	41.4	(21)
(22)			MCWB	35.2	35.0	42.1	49.0	60.6	67.9	70.2	69.5	67.1	60.1	48.8	39.4	(22)
(23)		5%	DB	34.4	34.8	43.2	55.5	69.7	77.5	80.3	78.2	73.8	62.0	49.2	38.7	(23)
(24)			MCWB	33.0	32.8	38.2	45.3	57.9	65.2	68.0	67.6	65.3	57.1	46.1	37.0	(24)
(25)		10%	DB	32.1	32.4	39.1	50.5	64.5	73.3	77.1	75.0	70.2	57.9	46.1	36.3	(25)
(26)			MCWB	30.8	30.6	35.4	42.9	54.5	62.9	66.5	65.7	63.4	54.0	43.6	34.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.3	39.7	50.5	56.3	68.0	73.7	75.3	73.8	71.8	65.7	54.3	44.4	(27)
(28)			MCD B	40.4	41.5	54.3	65.6	79.3	83.7	83.8	81.7	79.5	69.2	58.0	45.1	(28)
(29)		2%	WB	35.5	35.2	43.5	50.6	63.7	70.0	72.3	71.5	69.1	61.2	49.6	39.7	(29)
(30)			MCD B	36.7	37.1	47.7	57.8	72.2	78.8	80.4	78.0	75.0	65.7	52.0	40.8	(30)
(31)		5%	WB	33.2	33.1	38.8	46.6	59.4	67.1	70.0	69.4	66.9	57.6	46.3	37.2	(31)
(32)			MCD B	34.4	34.7	42.2	54.2	66.3	74.8	77.8	75.5	72.0	61.2	48.3	38.6	(32)
(33)		10%	WB	30.7	30.6	35.6	43.3	55.9	64.4	67.9	67.6	64.6	54.4	43.8	34.8	(33)
(34)			MCD B	32.1	32.3	38.8	49.5	63.1	70.9	74.9	73.3	69.2	57.8	46.0	36.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.2	16.6	17.7	17.3	20.9	21.3	20.3	19.1	18.3	14.4	11.5	12.2	(35)
(36)			MCDBR	11.7	14.4	19.5	28.0	29.8	27.9	26.3	23.8	22.4	20.2	15.6	12.2	(36)
(37)			MCWBR	11.1	12.9	13.8	17.4	17.5	16.1	14.3	13.3	13.1	14.0	12.6	11.3	(37)
(38)		5% WB	MCDBR	11.3	14.0	17.3	24.4	25.5	23.6	21.9	19.3	19.7	17.8	13.8	12.0	(38)
(39)			MCWBR	11.0	13.0	12.9	16.8	16.5	14.8	13.2	11.7	12.7	13.1	12.0	11.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.280	0.290	0.319	0.356	0.370	0.388	0.389	0.379	0.366	0.331	0.302	0.286	(40)	
(41)		taud	2.339	2.297	2.330	2.330	2.374	2.345	2.363	2.392	2.424	2.504	2.508	2.373	(41)	
(42)		Ebn at Noon	259	280	288	285	284	278	276	275	269	264	252	243	(42)	
(43)		Edh at Noon	23	30	34	37	37	38	37	35	31	25	20	21	(43)	
(44)	All-Sky Solar Radiation	RadAvg	340	632	1030	1411	1763	1901	1912	1617	1206	701	366	264	(44)	
(45)		RadStd	36	68	118	161	142	128	108	106	90	81	48	23	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page